Numerical Methods
for Stochastic Differential Equations

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Abstract

We examine numerical methods to simulate stochastic differential equations. Analytical solutions are presented and followed by numerical simulations.

Keywords: Ito Calculus, Stochastic Calculus, Timeseries, Monte-Carlo, Order-book Microstructure

Motivation

In most practical settings, stochastic differential equations (SDEs) cannot be solved analytically. In such cases, we turn to numerical methods to simulate a solution. We review the analytical solutions and then apply the numerical methods.

Ordinary Differential Equations

In ordinary calculus, we usually want to find values of a function given the rate of change. For example, let $f(t)$ denote the position of a particle at time t, and that we are given the function for the rate of change as

$$df(t) = C\big(f(t), t\big)dt$$

Then given the initial condition $f(0) = x_0$, we can approximate the function for some small increment of t using Euler's method

$$f(t + \Delta t) = f(t) + C\big(f(t), t\big)\Delta t$$

Euler's method is a simple way to approximate the ordinary differential equation. By taking smaller step sizes of Δt we could reach closer approximations to the exact value of the function. On the other hand, the method to approximate a stochastic differential equation is by using the **Euler-Maruyama** method.

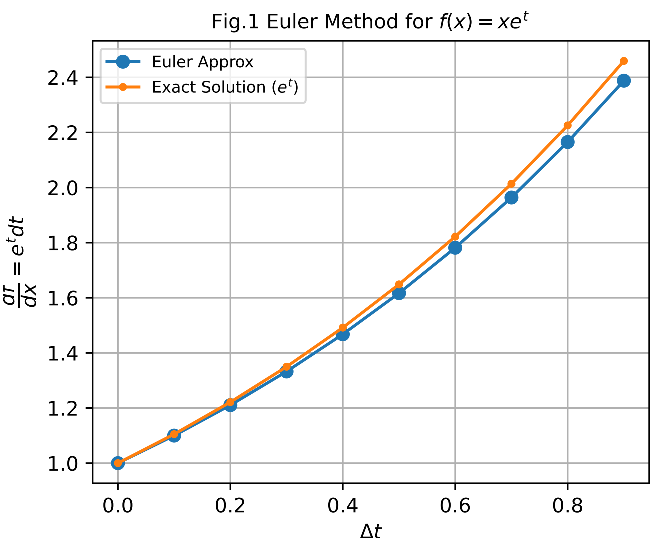

Figure 1: Euler Method

Stochastic Differential Equation

Euler-Maruyama(various authors,) method is derived by approximating the integral form of the stochastic differential equation over an infinitesimal time interval $[t_n, t_{n+1}]$. It is the stochastic counterpart of the standard Euler's method for ordinary differential equations. Consider the stochastic process $X(t)$ that satisfies the Ito differential equation

$$dX_t = f(X_t)dt + g(X_t)dW_t \tag{1}$$

where f is the drift coefficient and g is the diffusion coefficient, moreover W_t is a Brownian process. We express the given differential equation integrally as a single time step $\Delta t = t_{n+1} - t_n$,

$$X_{t_{n+1}} = X_{t_n} + \int_{t_n}^{t_{n+1}} f(X_s)ds + \int_{t_n}^{t_{n+1}} g(X_s)dW_s$$

then discretize the integral equation and assume that the coefficients $f(X_s)$ and $g(X_s)$ are constants over the interval $[t_n, t_{n+1}]$. We evaluate the integral from the left endpoint.

$$f(X_s) \approx f(X_{t_n})$$
$$g(X_s) \approx g(X_{t_n})$$

Substitute to the integral equation

$$X_{t_n+1} = X_{t_n} + f(X_{t_n})\int_{t_n}^{t_{n+1}} ds + g(X_{t_n})\int_{t_n}^{t_{n+1}} dW_s$$

Evaluate the deterministic integral $\int_{t_n}^{t_{n+1}} ds$ that simply turns to Δt. The stochastic integral $\int_{t_n}^{t_{n+1}} dW_s$ represents the increase of the Wiener processes denoted as $\Delta W_n = W_{t_{n+1}} - W_{t_n}$

Substitute the evaluated terms to (1)

$$X_{t_{n+1}} = X_n + f(X_n)\Delta_t + g(X_n)\Delta W_n$$

In practical simulations, ΔW_n is drawn from a normal distribution with mean 0 and variance Δt, often implemented as $\sqrt{\Delta t} \cdot \mathcal{N}(0, 1)$ Therefore, the approximation using Euler-Maruyama method is derived as

$$X_{t_{n+1}} = X_n + f(X_n)\Delta_t + g(X_n)\Delta W_n$$

This formula provides approximation for the depth of the path of the stochastic process by treating drift and diffusion as constants over each timestep. On the other hand, we compute the actual price S_t as a geometric Brownian process that changes by a random factor over a time interval Δt.

$$\Delta S = S(t + \Delta t) - S(t)$$

From the differential form

$$dS = \mu S dt + \sigma S dW_t$$

We use $S(t + \Delta t)$ below which is solved using Ito's lemma

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right]$$

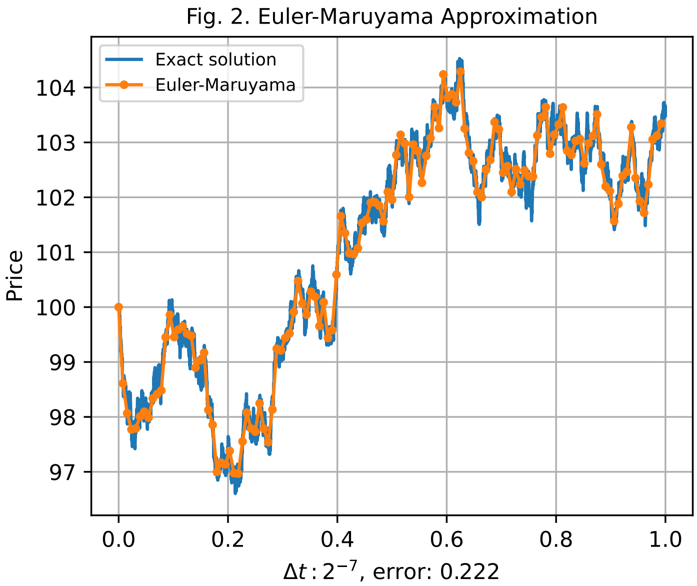

Figure 2: Euler-Maruyama Approximation

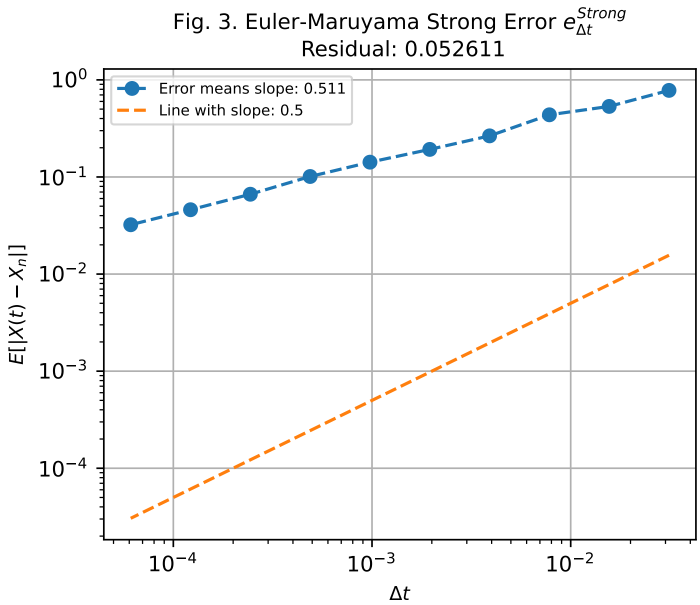

Figure 3: Euler-Maruyama Strong Error Convergence

Order of Convergence

Definition. Let $X(t)$ be a process and X_n denote the approximation of the value of $X(T)$ where $T = n\Delta t$. The weak error $e_{\Delta t}^{\text{weak}}$ with respect to some function f is defined as:

$$e_{\Delta t}^{\text{weak}} = \sup_{0 t_n \le T} \left| \mathbb{E}\big[f(X_n)\big] - \mathbb{E}\big[f(X(t_n))\big] \right|$$

We say that X_n converges weakly to $X(t)$ with order $\rho > 0$ if there exists a constant $C > 0$ independent of Δt such that:

$$e_{\Delta t}^{\text{weak}} \le C\Delta t^{\rho}$$

for all sufficiently small Δt.

Definition. We define the strong error as $e_{\Delta t}^{\text{strong}}$ as:

$$e_{\Delta t}^{\text{strong}} = \sup_{0 \le t_n \le T} \mathbb{E}\left[|X_n - X_{t_n}|\right]$$

We can say X_n converges strongly to X_t with the order $\rho > 0$ if there exists a constant $C > 0$ independent of Δt such that

$$e_{\Delta t}^{\text{strong}} \le C\Delta t^{\rho}$$

for all sufficiently small Δt. Strong order convergence is usually lower than the weak order convergence since it is more restrictive and demands more accuracy for each timestep while weak convergence is less restrictive where each path can diverge as long as distribution is more controlled.

Euler-Maruyama method converges strongly with the order $\rho = \frac{1}{2}$ or $O(\sqrt{\Delta t})$. This means that error decreases proportional to the square root of the step size, assuming the coefficients of the SDE satisfy Lipschitz(various authors,) continuity condition. We test this assumption by taking the mean of errors from time steps $\Delta t = 2^{-x}$ for values $5 < x < 15$(Kalingking,). Taking the log of both sides

$$log\left(e_{\Delta t}^{\text{strong}}\right) \le log(C) + \frac{1}{2} log(\Delta t)$$

The experimental slope of the error means shown in Fig. 3 is close identical to the sample line having a slope equal to $\frac{1}{2}$. This proves numerically the strong error, $e_{\Delta t}^{\text{Strong}}$, converges in the order of $\rho = \frac{1}{2}$.

We can view weak convergence as the decay in the rate of the "error of the means" while strong convergence is the decay in the rate of the "mean of the errors". Strong convergence implies weak convergence, but the opposite is not necessarily true.

If a numerical scheme converges with the order ρ, then decreasing the step size by a factor of k reduces the error by a factor of k^{ρ}. For example, the order of $\rho = 1$ means that decreasing the step size by a factor of 100 also decreases the error by a factor of 100. Meanwhile, an order of $\rho = \frac{1}{2}$ means that decreasing the step size by a factor of 100 decreases the error by a factor of 10.

We also assume that Euler-Maruyama approximation converges weakly to the exact solution $X(t)$ at time T if for a class of functions the error in the expected value vanishes as Δt approaches 0. As such, this numerical scheme focuses on statistical properties (e.g. mean and variance) of multiple simulation paths rather than a single solution.

$$e_{\Delta t}^{\text{Weak}} := \big|E[X_n] - E[X(T)]\big| \Rightarrow \ 0 \ as \ \Delta t \Rightarrow 0$$

The order of convergence ρ is generally equal to 1 which means the error decreases linearly as Δt goes to zero.

$$e_{\Delta t}^{\text{Weak}} \le C\Delta t^{\rho}$$

To test this hypothesis, let us first find $E[X(t)]$, recall that:

$$dX_t = \mu X_t dt + \sigma X_t dB_t$$

Converting to integral form:

$$X_t = X_0 + \mu \int_0^t X_s ds + \sigma \int_0^t X_s dB_s$$

We take the expectation of both sides

$$\mathbb{E}[X_t] = \mathbb{E}[X_0] + \mu\mathbb{E}\left[\int_0^t X_s ds\right] + \sigma\mathbb{E}\left[\int_0^t X_s dB_s\right]$$
$$d\mathbb{E}[X_t] = \mathbb{E}[X_0] + \mu\mathbb{E}[X_t]ds + 0$$

This implies

$$d\mathbb{E}[X_t] = \mu\mathbb{E}[X_t]dt, \quad \mathbb{E}[X_0] = x_0$$

which solves to a known solution

$$d\mathbb{E}[X_t] = X_0 e^{ut}$$

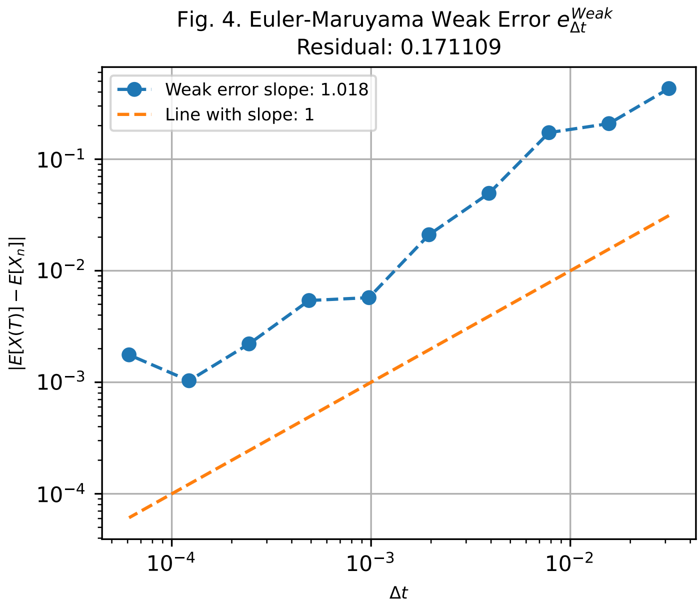

Figure 4: Euler-Maruyama Weak Error Convergence

We run the numerical experiment with $M = 10^6$ sample paths using Monte-Carlo simulation with values of $\Delta t = 2^{-x}$ such that $5 < x < 11$

$$\left|\mathbb{E}[X_n] - \mathbb{E}[X(T)]\right|, \quad n\Delta t = T$$

The experimental slope of the error means closely matches the weak convergence order of $\rho = 1$. The slope of the errors in Fig. 4 is visually identical to the sample line with slope equal to 1.

Milstein Method

The **Milstein**(various authors,) method improves the Euler-Maruyama approximation by adding the second-order term from the Ito-Taylor expansion. This increases the strong convergence order from 0.5 to 1.0.

Consider a stochastic differential equation of the form as shown in (1) The Euler-Maruyama method only keeps the first order terms while Milstein includes an additional term to account for the noise on diffusion signal. Apart from identifying the drift $a(X_t)$ and the diffusion $b(X_t)$, we also need the derivative $b'(X_t)$.

$$\begin{aligned} f(x) &= \mu x \\ g(x) &= \sigma x \\ g^{'}(x) &= \sigma \end{aligned}$$

We derive the Milstein method from the diffusion term of (1) by performing the truncated Taylor expansion of $g(X_t)$ around X_n using Ito's lemma

$$g(X_t) \approx g(X_n) + g^{'}(X_n)(X_t - X_n)$$

Substitute $g(X_t) \approx g(X_n)dW_t$ into $g(X_t)dW_t$ integral

$$\int_{t_n}^{t_{n+1}} g(X_s, s)dW_s$$

becomes

$$\int_{t_n}^{t_{n+1}} g(X_t)dW_s + \int_{t_n}^{t_{n+1}} g^{'}(X_t, t)(X_s - X_t)dW_s$$

We approximate $(X_s - X_t)$ from the diffusion term of (1) to capture the stochastic behavior

$$X_s - X_t \approx f(X_t, t)(s - t) + g(X_t, t)(W_s - W_t)$$

and substitute to the second integral

$$\begin{aligned} \int_{t_n}^{t_{n+1}} g(X_t)dW_t &\approx \int_{t_n}^{t_{n+1}} \left[g(X_n) + g^{'}(X_n)g(X_n)(W_{t_{n+1}} - W_{t_n})\right] dW_t \\ &\approx \int_{t_n}^{t_{n+1}} g(X_n)dW_s \\ &\qquad + \int_{t_n}^{t_{n+1}} g(X_n)b^{'}(X_n)(W_s - W_{t_n})dW_s \\ &\approx g(X_n)\Delta W_n + g(X_n)g^{'}(X_n)\int_{t_n}^{t_{n+1}} (W_s - W_{t_n})dW_s \end{aligned}$$

To solve the remaining integral, we use Ito's formula on the process $f(x) = x^2$ applied to $X_s := W_s - W_{t_n}$ for all $s \in [t_n, t_{n+1}]$ We now use $f(x) = f(W_s) = (W_s - W_{t_n})^2$ and by Ito's formula

$$df(W_s) = f'(W_s)dW_s + \frac{1}{2}f"(W_s)dW_s^2$$

Compute the derivatives

$$\begin{aligned} f'(W_s) &= 2(W_s - W_t) \\ f^{"}(W_s) &= 2 \\ (dW_s)^2 &= ds \end{aligned}$$

Substitute the derivatives to Ito's formula

$$\begin{aligned} d[(W_s - W_t)^2] &= 2(W_s - W_t)dW_s + \frac{1}{2}(2)ds \\ &= 2(W_s - W_t)dW_s + ds \end{aligned}$$

Integrate from t to s

$$\begin{aligned} \int_t^s d(W_s - W_t)^2 &= \int_t^2 2(W_s - W_t)dWs + \int_t^2 ds \\ (W_s - W_t)^2 - (W_t - W_t)^2 &= 2\int_t^s (W_s - W_t)dW_s + (s - t) \\ (W_s - W_t)^2 &= 2\int_t^s (W_s - W_t)dW_s + (s - t) \end{aligned}$$

Rearranging to find the integral

$$\begin{aligned} \int_t^s (W_s - W_t)dW_s &= \frac{1}{2}\left[(W_s - W_t)^2 - (s - t)\right] \\ &= \frac{1}{2}(dW_t)^2 - dt \end{aligned}$$

Combining this extra correction term with the original Euler-Maruyama expression, we get the Milstein method with the added correction term.

$$X_{t_{n+1}} = X_t + f(X_t)dt + g(X_t)dW_t + \frac{1}{2}g'(X_t)g(X_t)\left[(dW_t)^2 - dt\right]$$

This term specifically targets the SDE's where the diffusion $g(X_t)$ depends on the state X_t. If $g(X_t)$ is constant, the derivative $g'(X_t)$ is zero and Milstein method collapses back to Euler-Maruyama. We compare the performance of the Milstein method in Fig. 5 with Euler-Maruyama in Fig. 3 with the same Monte-Carlo simulation size and we can confirm improvement of the slope of the means of error. The following Fig. 6 illustrates the Milstein weak error order of 1 which is normally used where overall distribution is more important when assessing the reliability of the numerical method.

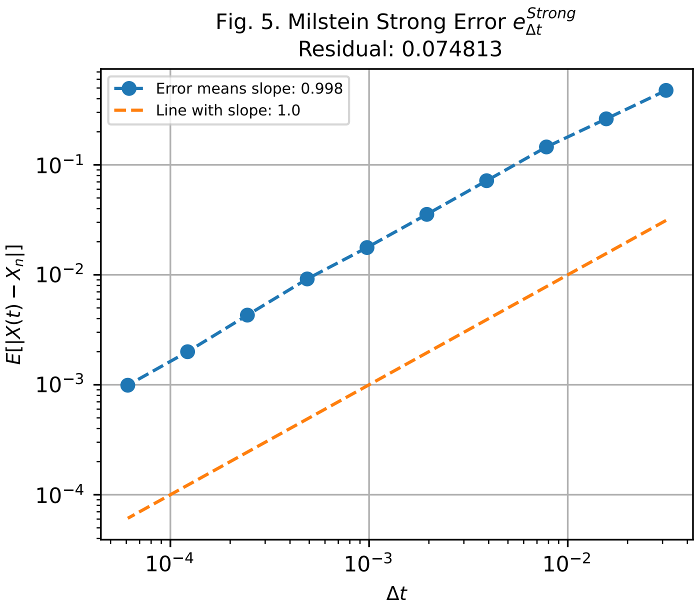

Figure 5: Milstein Strong Error Convergence

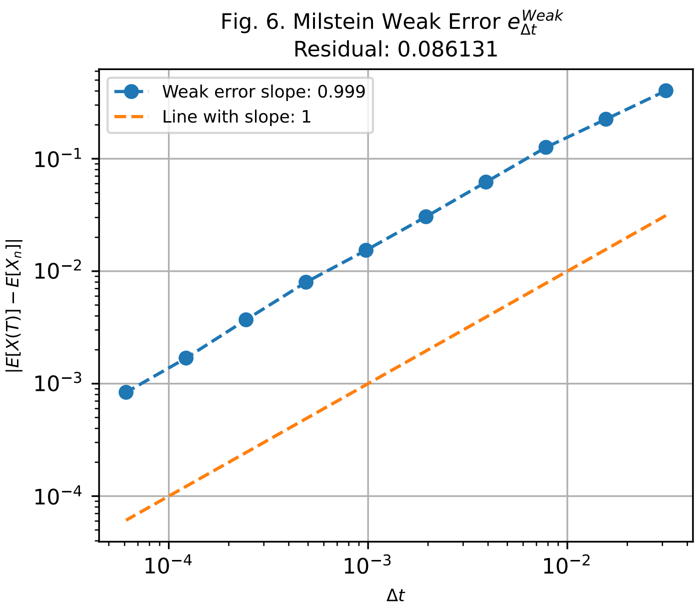

Figure 6: Milstein Weak Error Convergence

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